

Course End Project- Enterprise Infrastructure Security

Enhancing Windows Security with Logon Monitoring and Reporting

By

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1. Introduction:

Cyber threats targeting Windows-based systems are particularly through unauthorized login attempts and malicious command executions. This project focuses on enabling detailed audit logging, analyzing security events using Event Viewer and PowerShell, and generating automated HTML-based security reports. The goal is to proactively detect suspicious activities, improve visibility into user behavior, and strengthen the organization's overall security posture.

2. Problem Statement:

TechSecure, a mid-sized enterprise, faces increasing cyber threats targeting its Windows-based systems, particularly through unauthorized login attempts and malicious command executions.

The challenge is to:

- Analyze failed logins to identify every intrusion.
- Document every executed command.
- Real time tracking of successful logons.
- Create Reports of failed login attempt.

3. Project Objective:

To enhance Windows security by implementing advanced audit configurations, including failed login analysis, command logging, and real-time reporting. This project leverages Windows Audit configuration, Group Policies, Event Viewer, PowerShell automation and HTML based reporting to monitor, detect and mitigate potential security breaches.

4. Scope of the Project:

- Set up audit policy to do logon failure analysis.
- Configure command and process logging.
- PowerShell Logging Configuration
- Create a web-based reporting tool.

5. Implementation Details:

5.1. Advanced Audit Policy Configuration

Audit policies are configured using Microsoft Management Console (MMC) and Local Group Policy. Both successful and failed events are enabled to ensure complete visibility into authentication attempts, system access, and process execution.

Local Computer Policy → Computer Configuration → Windows Settings → Security Settings → Advanced Audit Policy Configuration

The following audit categories were configured for both **Success and Failure** events:

- **Account Logon**
 - Audit Credential Validation
 - Audit Other Account Logon Events
- **Logon/Logoff**
 - Audit Logon
 - Audit Logoff
 - Audit Special Logon
 - Audit Other Logon/Logoff Events
- **Detailed Tracking**
 - Audit Process Creation
 - Audit Process Termination
- **Object Access**
 - Audit File System
 - Audit Registry
- **Privilege Use**
 - Audit Sensitive Privilege Use
 - Audit Non-Sensitive Privilege Use
- **System**
 - Audit System Integrity
 - Audit Other System Events

Global Object Access Auditing was configured for both **File System and Registry**, applying auditing rules to the *Everyone* group.

5.2 PowerShell Logging Configuration

To ensure visibility into command execution and script activity, PowerShell logging was enabled via Group Policy:

- Turn on Script Execution → Enabled (Allow all scripts)
- Turn on Module Logging → Enabled (* to log all modules)

This ensures that all PowerShell activity is logged and can be analyzed for malicious behavior.

5.3. Event Generation and Validation

To validate the audit configuration, a new local user account named Bob was created. The following actions were performed:

- Execute the following commands as Bob:
- ping www.example.com
- Launching applications (e.g., MSPaint)
- Exit from user Bob.

These actions generated security events in Event Viewer, including:

- Logon events (Event ID 4624)
- Process creation and termination events (Event ID 4688, 4689)

Event Viewer filters were used to verify the presence of Event IDs such as 4688 and 4689, confirming that auditing was functioning correctly.

5.4 Security Event Analysis Using PowerShell

PowerShell scripts were used to extract and analyze specific security events from the Windows Security log.

Event ID 4624 : Successful logon

Event ID 4625 Failed logon attempt

Event ID 4688 Process creation

Event ID 4689 Process termination

A custom PowerShell script was developed to:

- Monitor recent security events
- Identify successful and failed login attempts
- Extract usernames and source IP addresses from failed logins
- Track executed processes and commands

This automation significantly reduces manual log analysis effort and enables faster detection of suspicious activity.

5.5 Automated HTML Security Report Generation

1. IIS Web Server Deployment

Internet Information Services (IIS) was installed to host security reports. The PowerShell script periodically collects security events and converts them into a structured HTML report stored in:

C:\inetpub\wwwroot\security_report.html

2. Real-Time Monitoring

The script runs continuously and updates the report every five minutes, providing near real-time security visibility. The report includes:

- Successful login activity
- Failed login attempts with source IP addresses
- Process creation and command execution events

The report is accessible via a web browser using:

- http://localhost/security_report.html
- http://localhost:8080/security_report.html (via port proxy configuration)

Firewall rules and port forwarding were configured to allow secure access to the report.

5.6. Test Implementation and Results

The implemented solution provides TechSecure with:

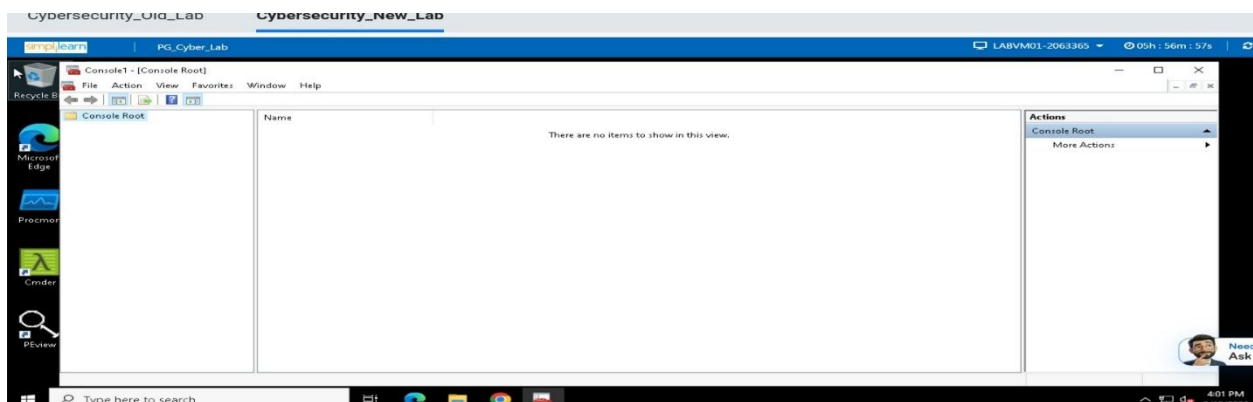
- Improved detection of **unauthorized login attempts**
- Visibility into **suspicious IP addresses**
- Comprehensive logging of **commands and processes**
- Automated, centralized security reporting
- Reduced response time to potential security incidents

This proactive monitoring approach helps prevent unnoticed intrusions and supports forensic investigations when incidents occur.

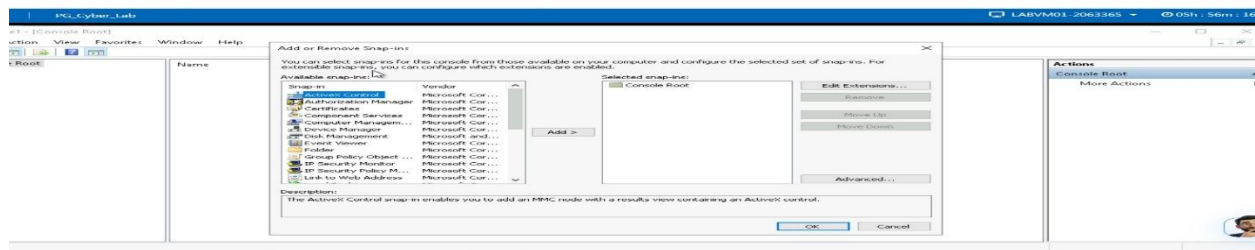
6. Steps to be followed to achieve the results:

Task 1: Set up audit policy to do logon failure analysis

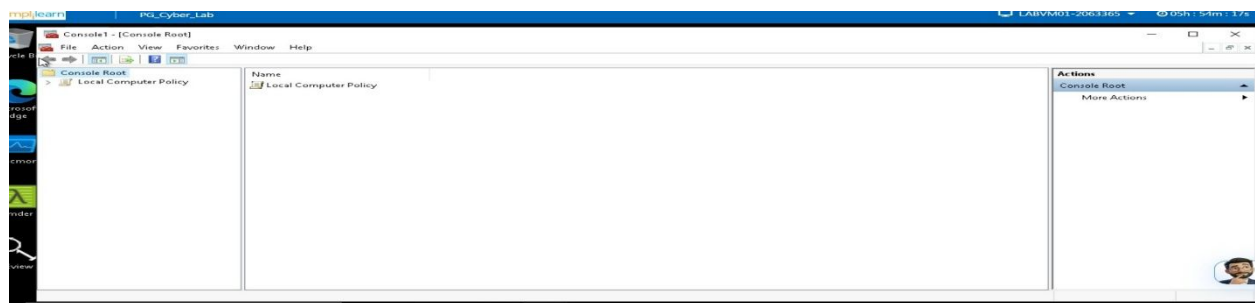
1.1 Open lab VM 01 > Windows. Type “Run”. Run Command opens up. Type “mmc” on run command line to open Microsoft Management Console.



1.2. Click on File > then click Add /Remove Snap in. Add Remove snap in opens up.



1.3 Select Group Policy Object Editor and Click Add button. Click Finish and followed by clicking OK button. Then the Local Computer Policy is added to the Microsoft Management Console.

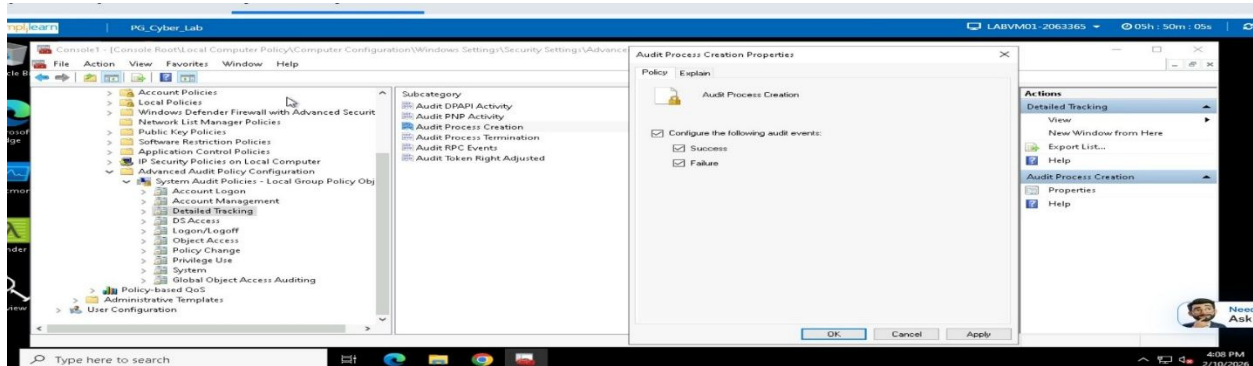
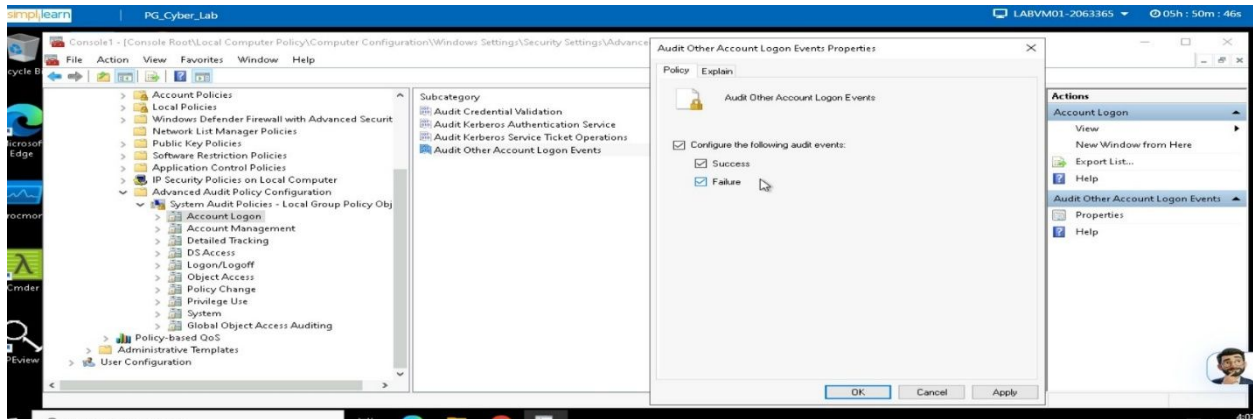
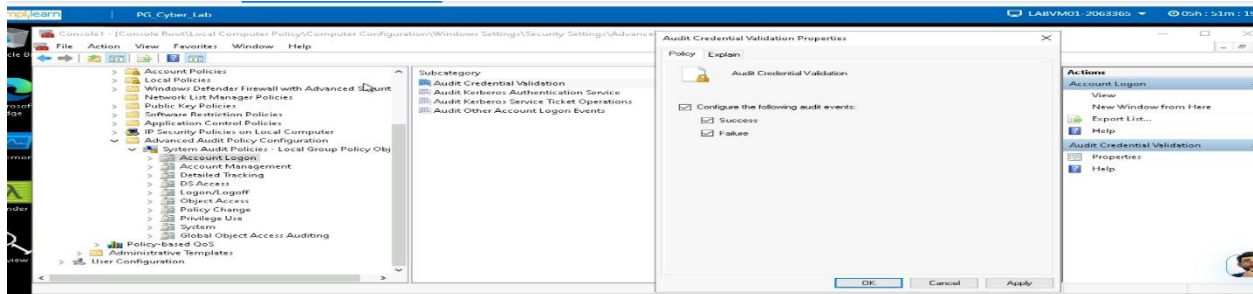
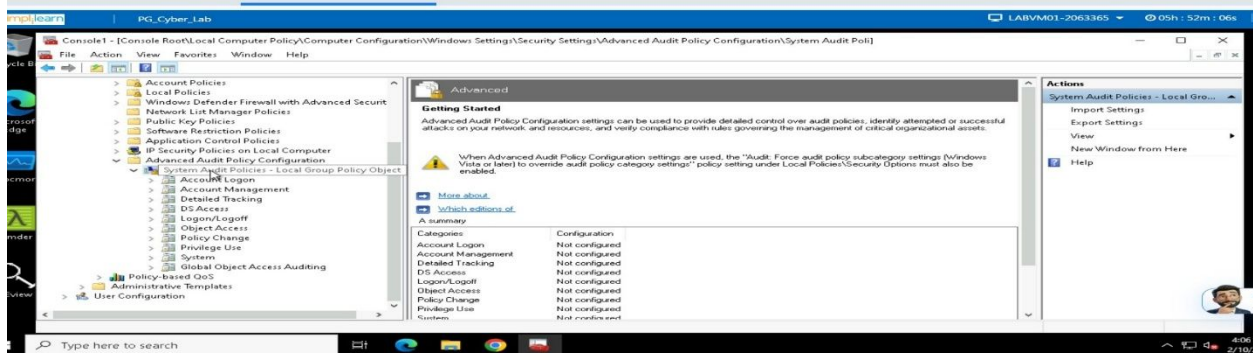


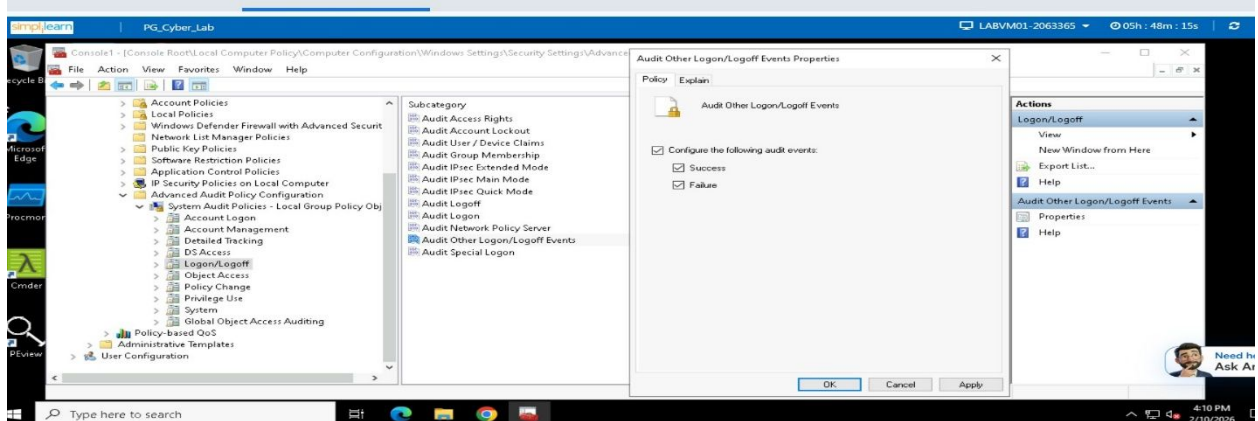
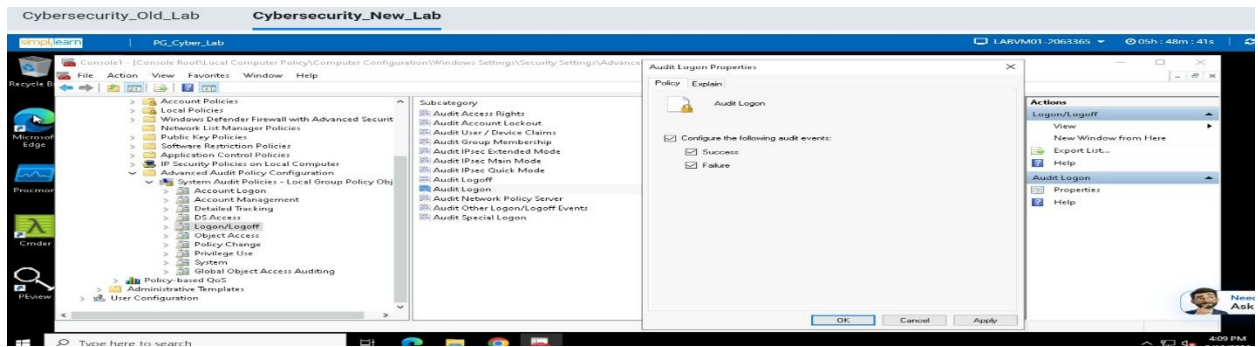
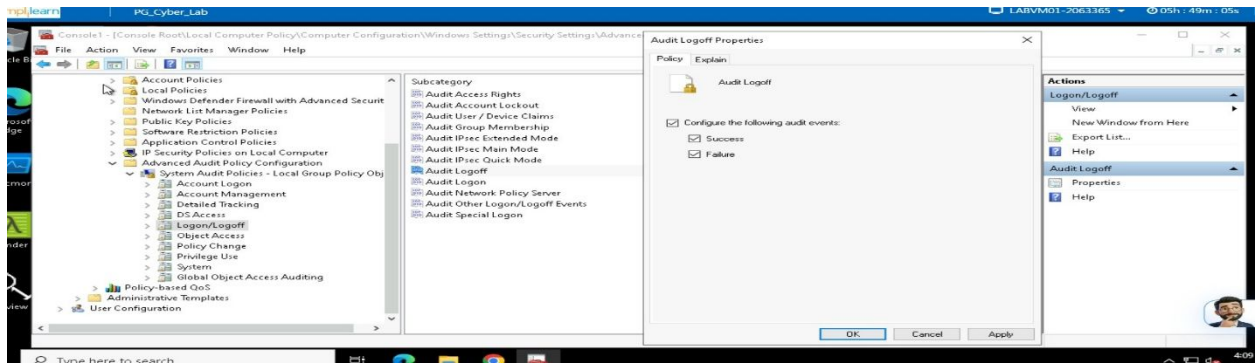
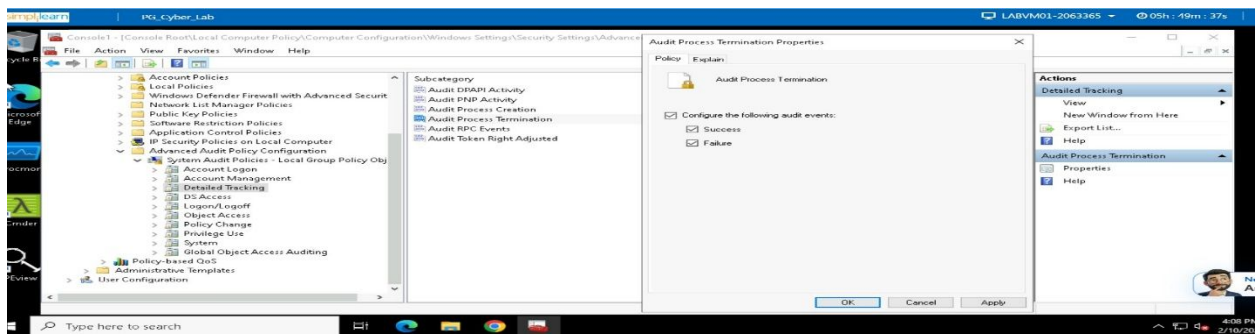
1.5. Click to Open > Local Computer Policy > Windows Settings > Security Settings > Advanced Audit Policy Configuration > System Audit Policies -Local Group Policy Object

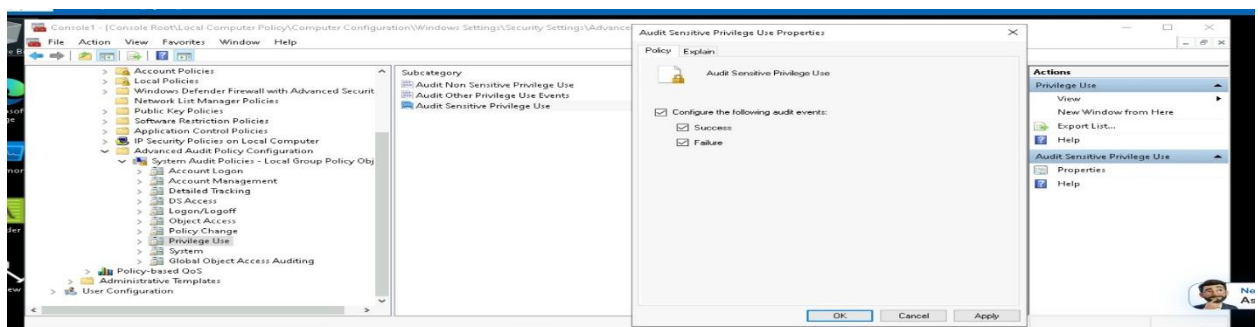
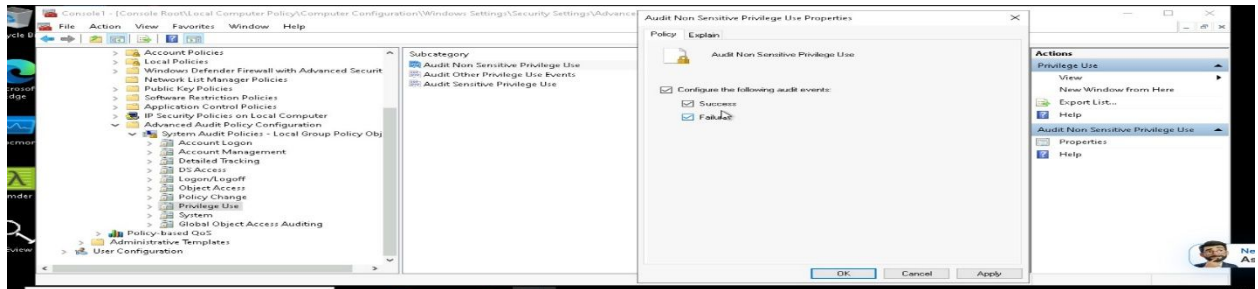
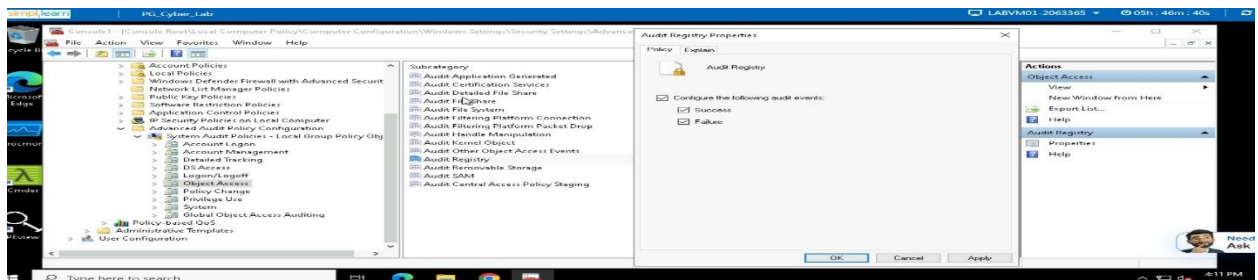
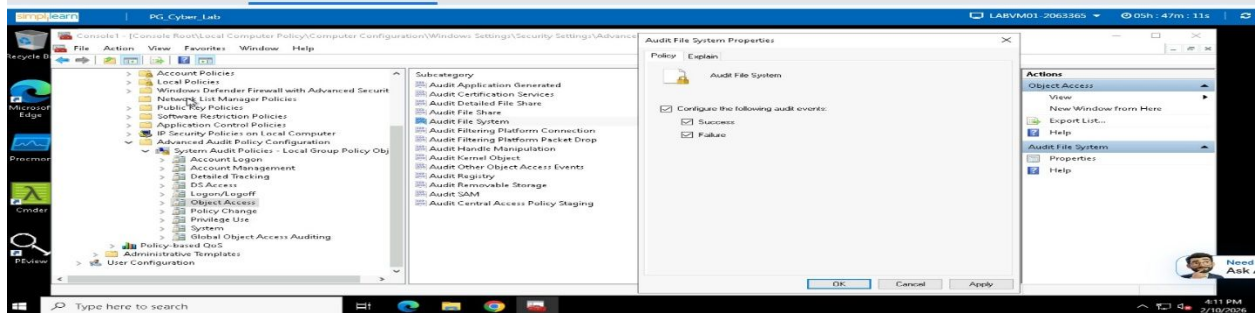
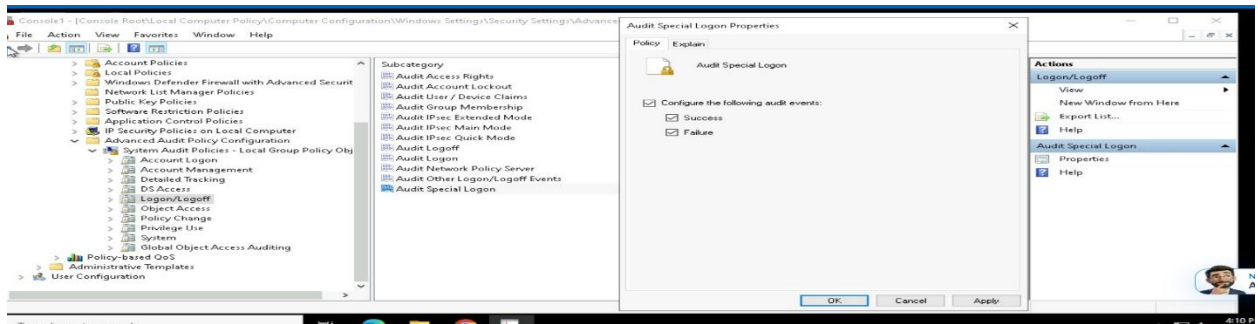
The following audit categories were configured for both **Success** and **Failure** events:

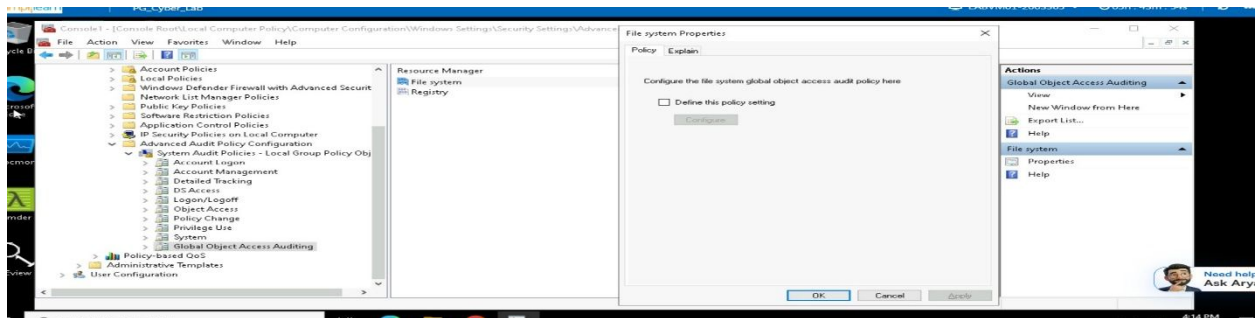
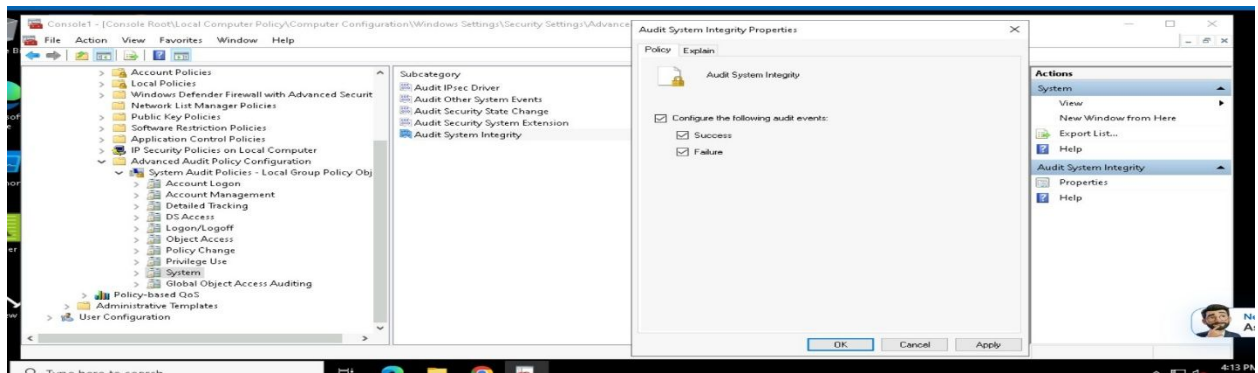
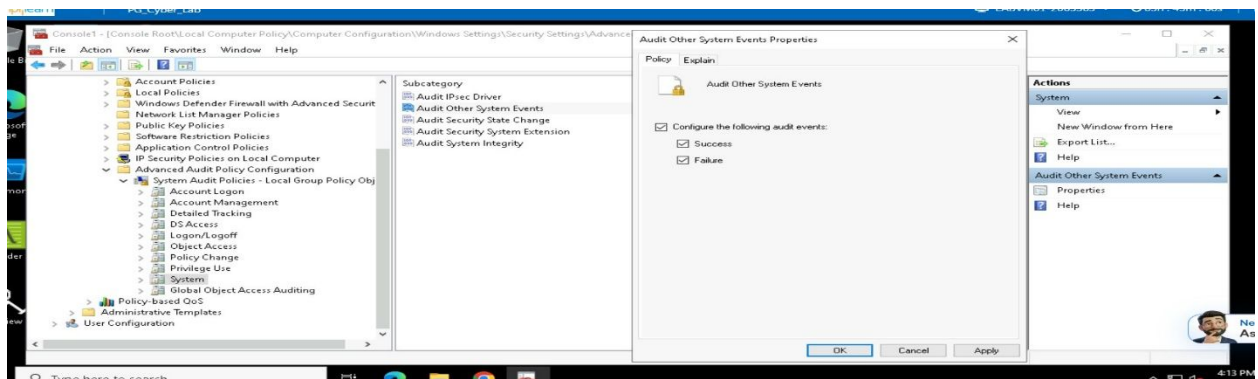
- **Account Logon**
 - Audit Credential Validation
 - Audit Other Account Logon Events
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 - Audit Non-Sensitive Privilege Use
- **System**

- Audit System Integrity
- Audit Other System Events

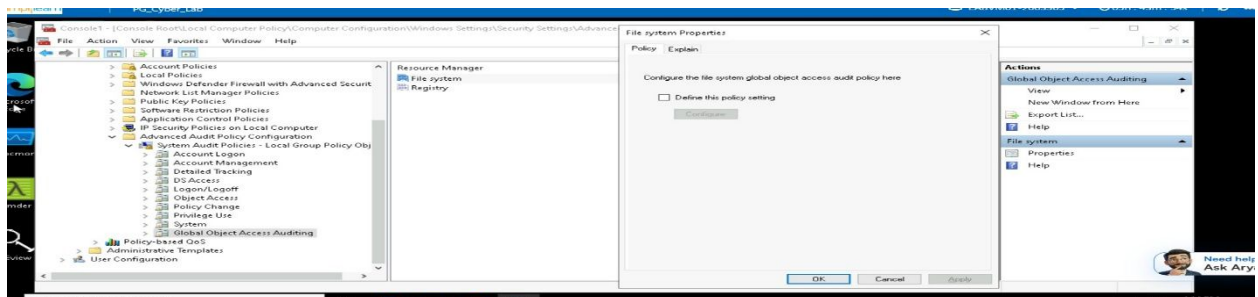


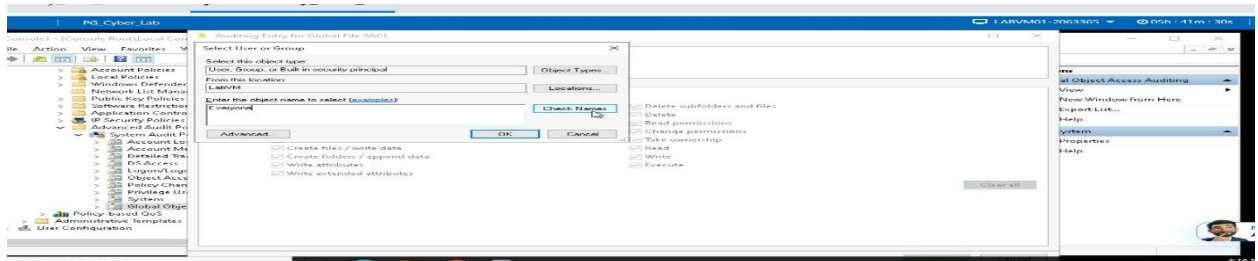
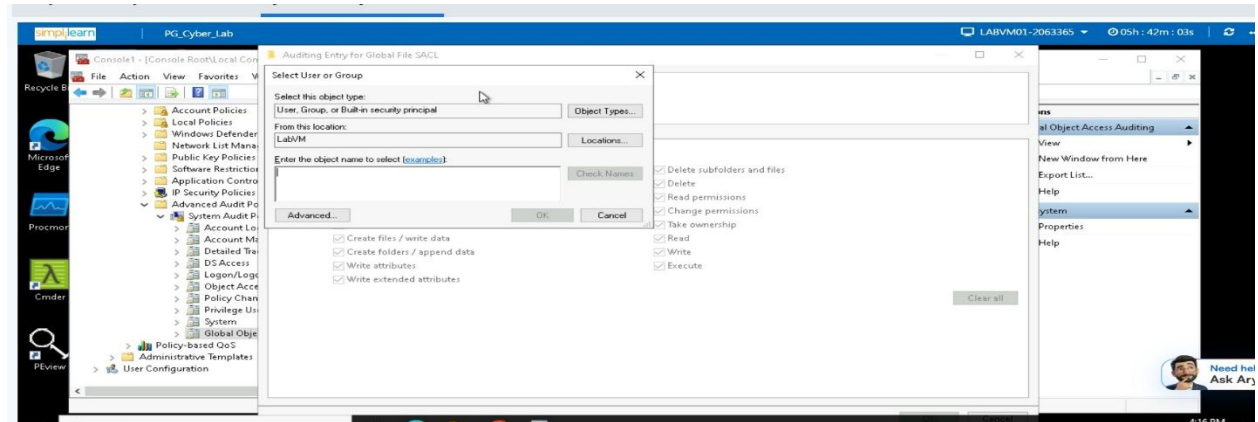
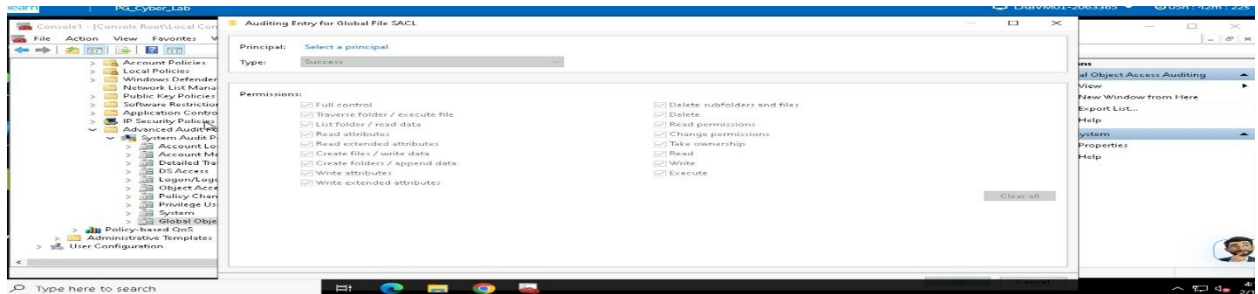
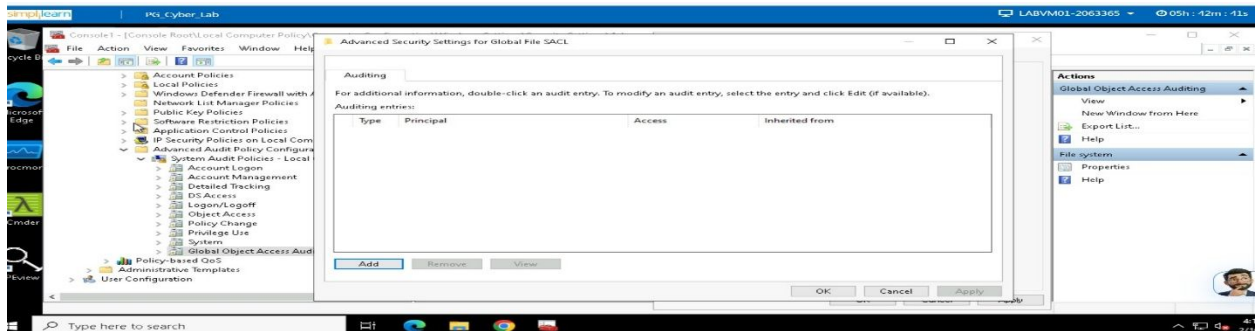


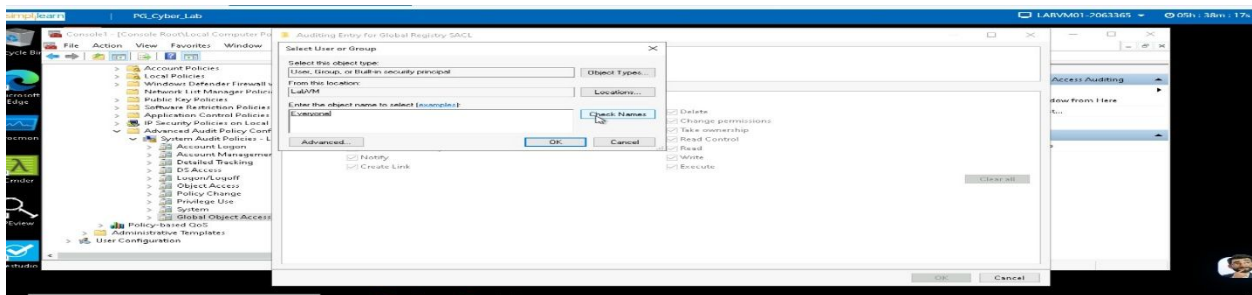
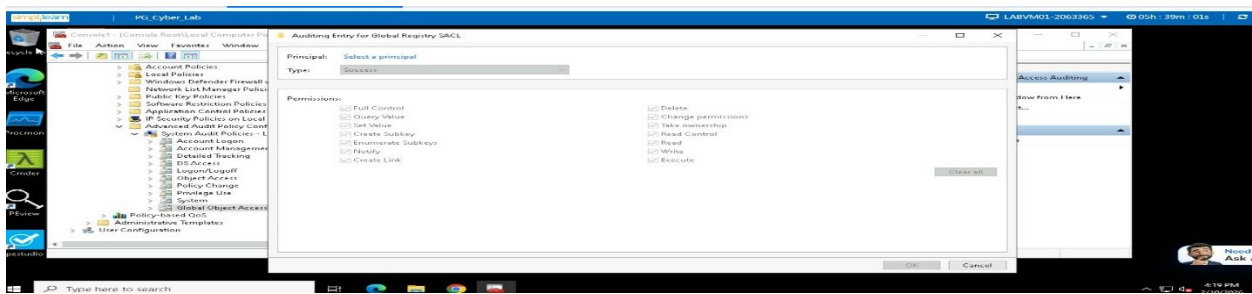
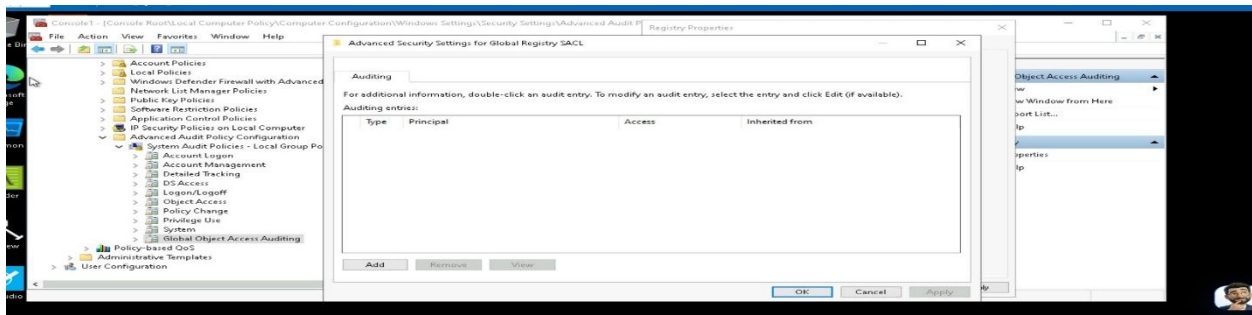
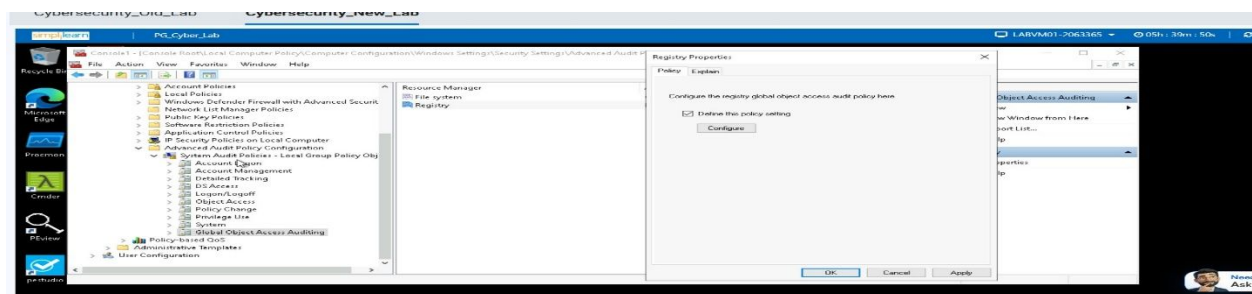
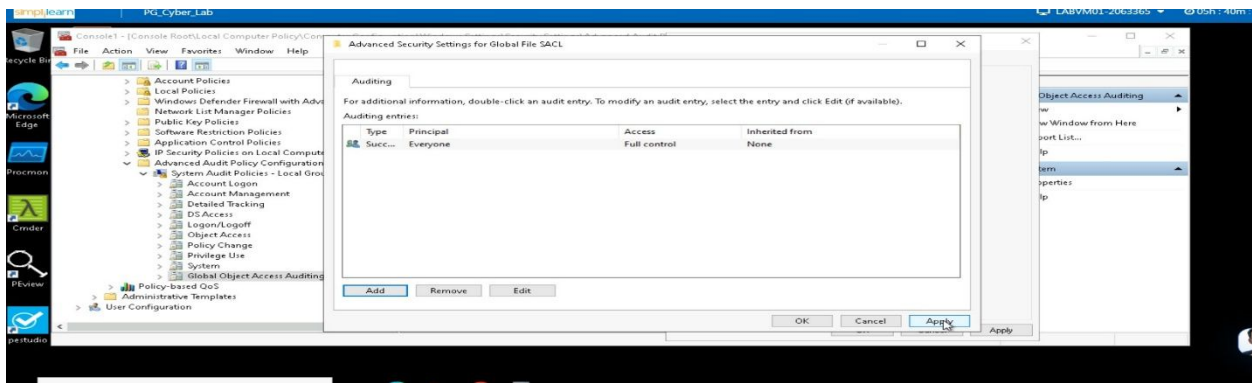


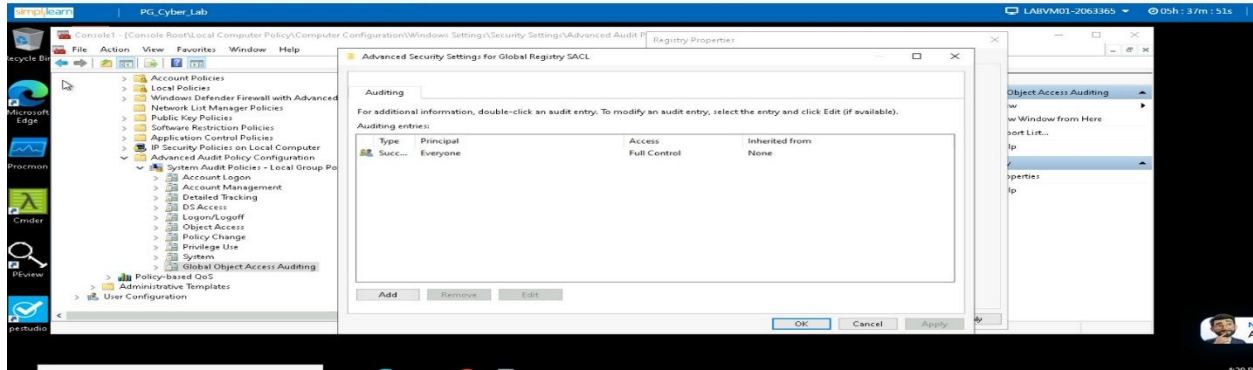


1.6 Click to Open > Global Object Access Auditing > Configure both **File System** and **Registry**. Click on Configure > Add> Select a principal > applying auditing rules to the *Everyone* group.



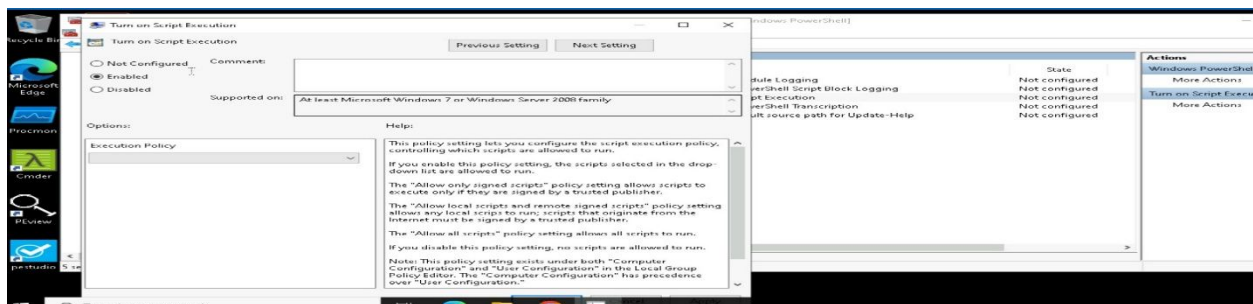
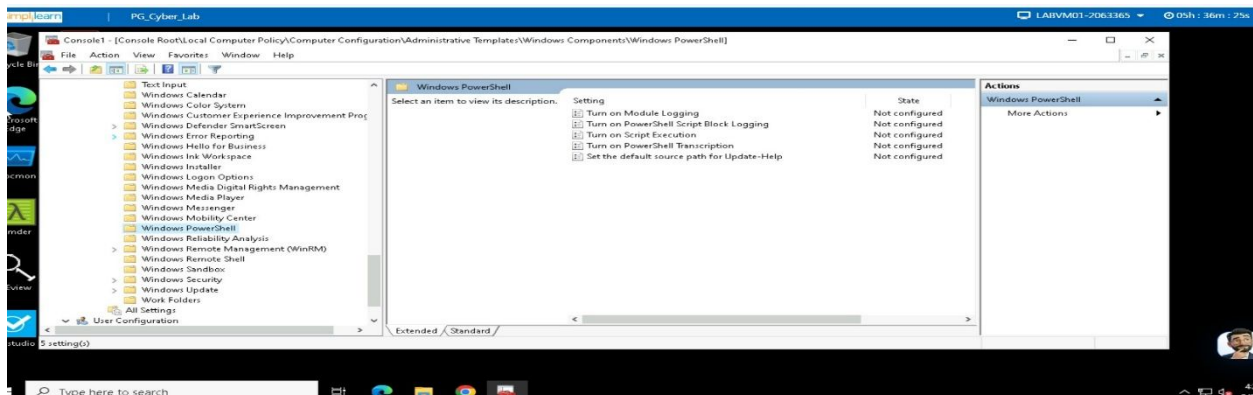


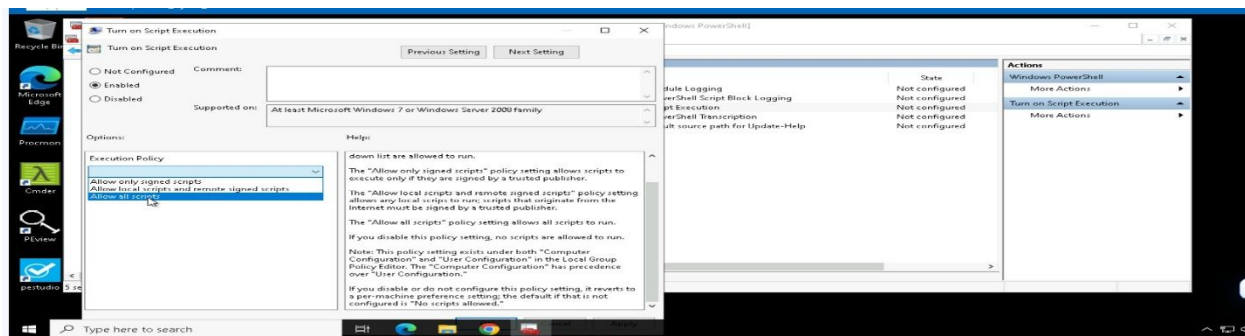




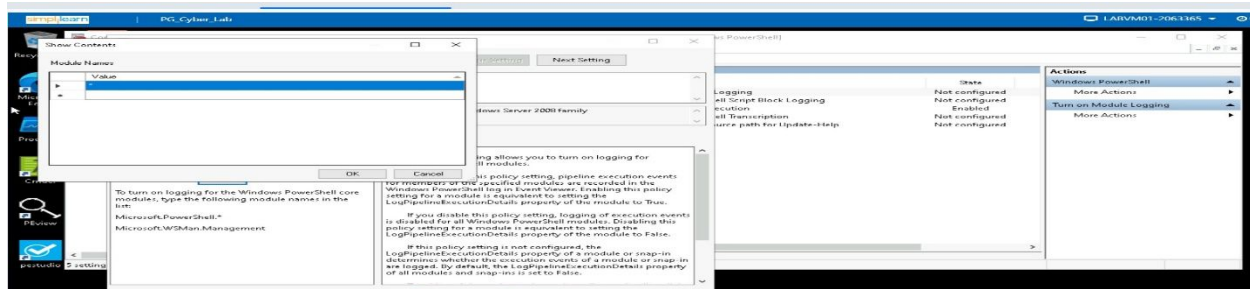
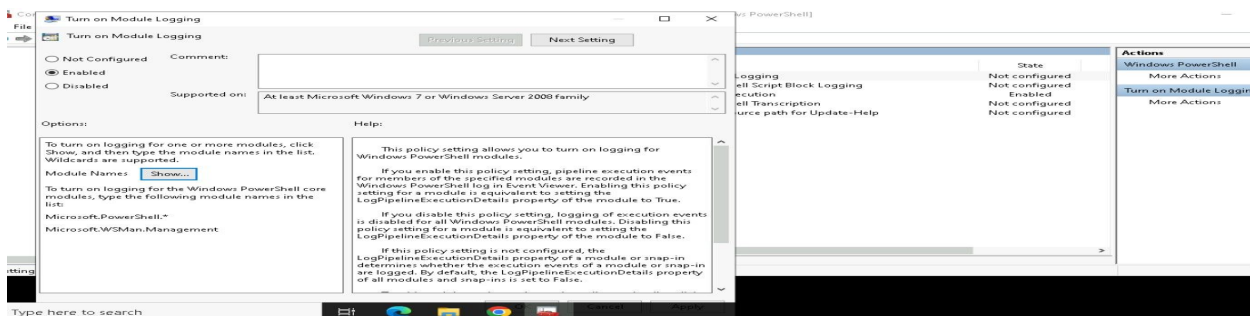
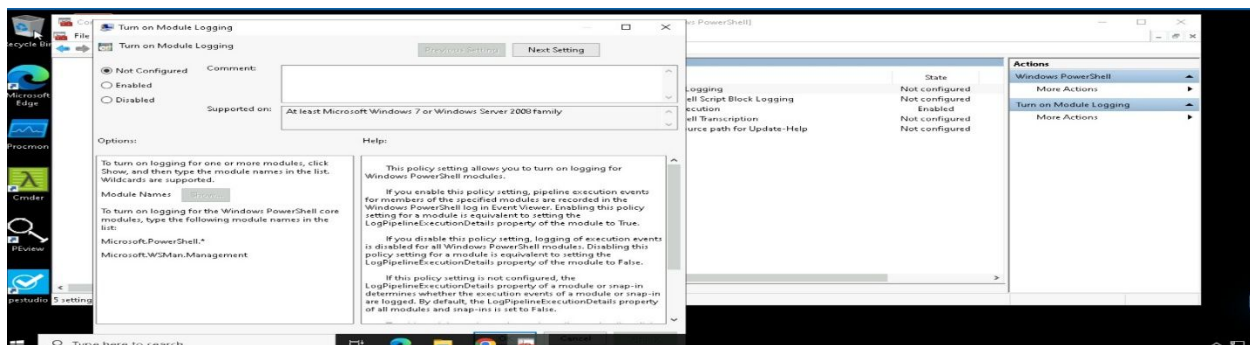
Task 2: PowerShell Logging Configuration

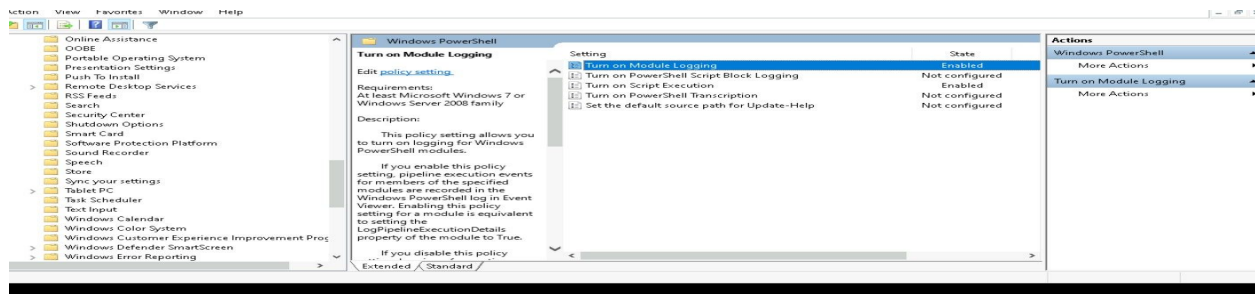
2.1 Click to Open > Navigate to Local Computer Policy > Computer Configuration > Administrative - Templates > Windows Components – Windows Powershell > -click Turn on Script Execution - Check the Enable button, and in the Execution Policy, select Allow all scripts then click on Apply and OK





2.2 Click to Open > Navigate to Local Computer Policy > Computer Configuration > Administrative - Templates > Windows Components – Windows Powershell > click Turn on Module Logging – Enable – Show - Type asterisk(*) under Value and click on OK – Apply – OK





Task 3: Event Generation and Validation

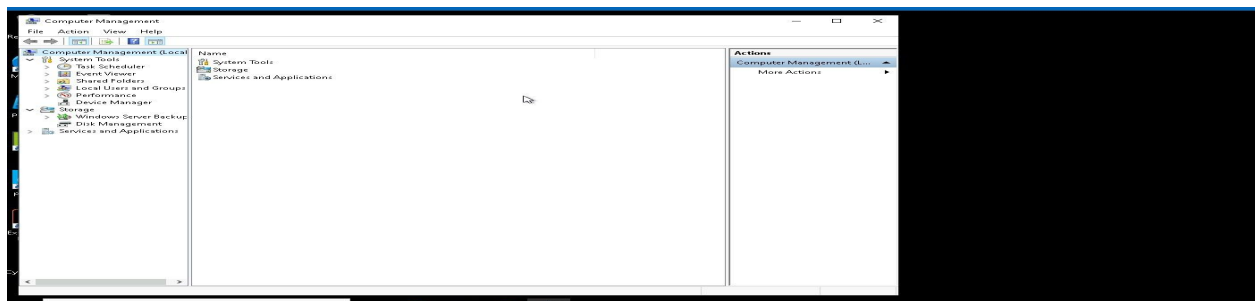
3.1 Open PowerShell as Administrator > Enter the below command to update the policy changes made in Task 1 and 2.

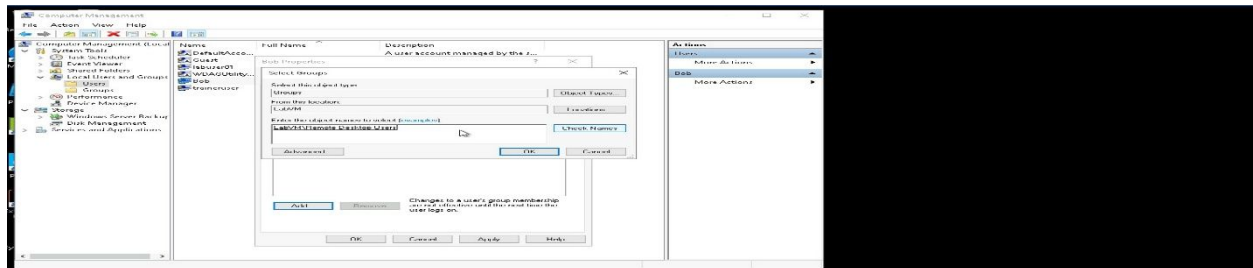
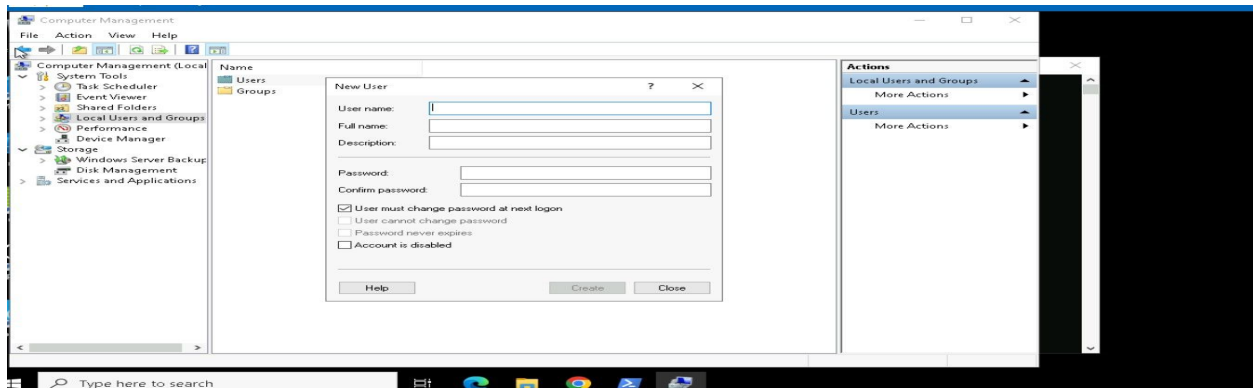
gpupdate /force



3.2 Create a user new local user

Open Computer Management System Tools > Local Users and Groups then right-click on User and click on New User... Bob as the User name, Bob Kumar as the Full Name, set a Strong Password, uncheck all options, and click Create . Add the created user to “Remote Desktop users” group.





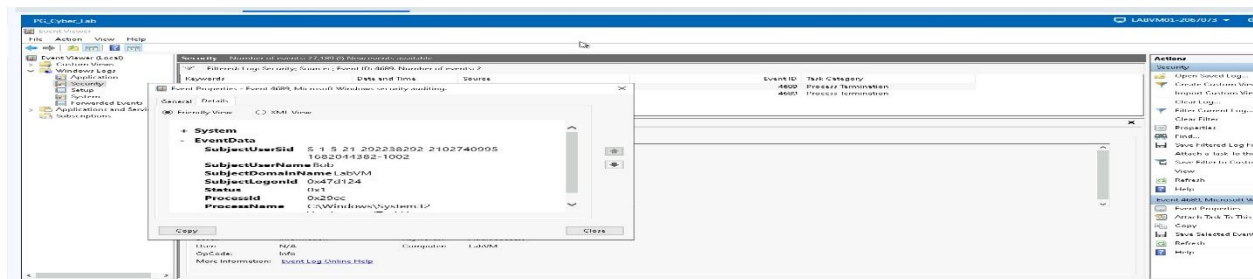
3.3 Generate Events as new local user.

VM machine does not allow user logins. In order to login as previously created user “Bob”, We need to use remote desktop connection on your machine. **Unable to capture the Screenshots from remote desktop connection.** Please follow the below steps.

1. Login to remote desktop connection as Bob
2. Open PowerShell and Run as different user.(Event ID 4624).
3. Enter the command ping www.example.com
4. After successful ping, enter mspaint to start a process creation (Event ID 4688).
5. Close the mspaint for process termination (Event ID 4689).
6. Navigate back and login back as Administrator.

3.4 Verify Events in Event Viewer.

Open Event Viewer Windows Logs and click on Security. click on Filter Current Logs. Enter the Event ID 4689 and click on OK



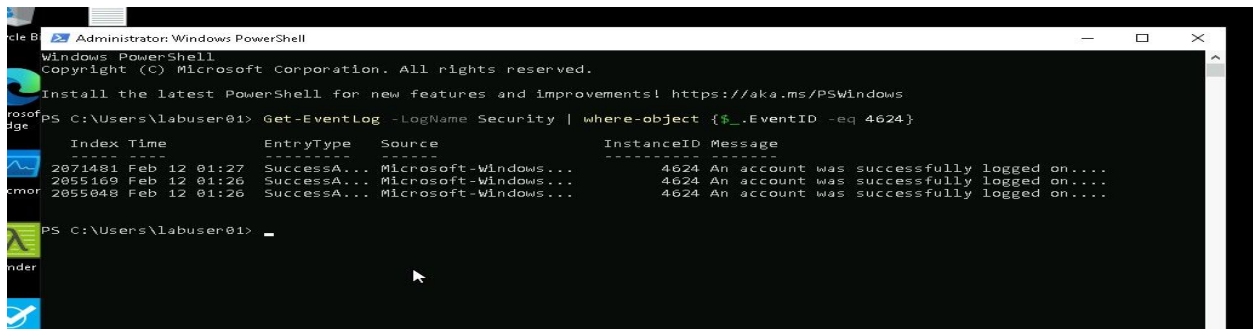
Task 4: Security Event Analysis Using PowerShell

PowerShell scripts were used to extract and analyze specific security events from the Windows Security log

4.1 Open PowerShell as Administrator.

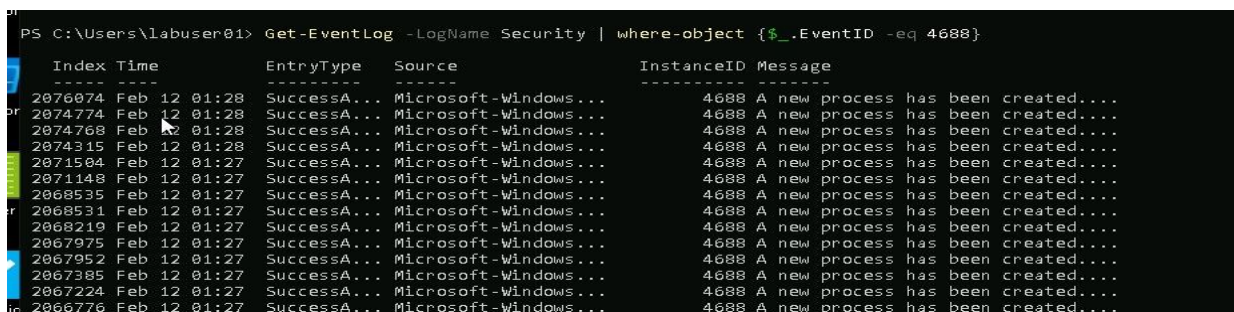
Enter the below commands to verify the events:

- `Get-EventLog -LogName Security | where-object {$_.EventID -eq 4624}`
- `Get-EventLog -LogName Security | where-object {$_.EventID -eq 4688}`
- `Get-EventLog -LogName Security | where-object {$_.EventID -eq 4689}`



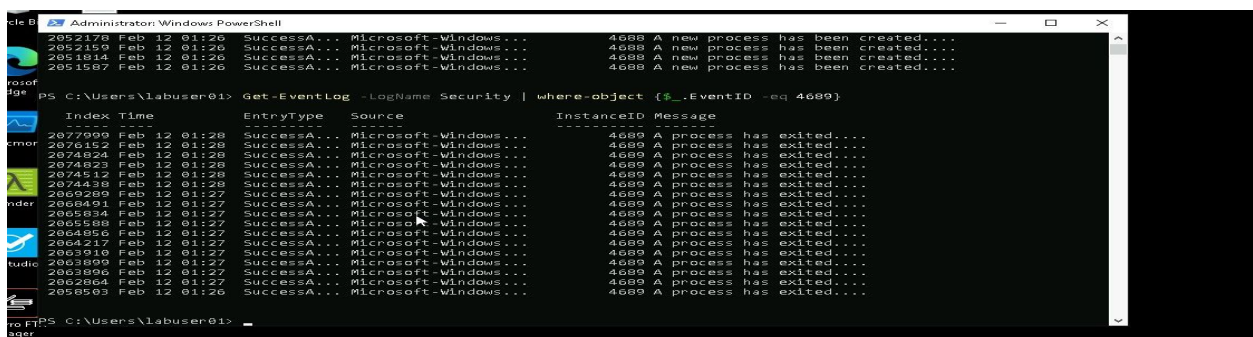
```
PS C:\Users\labuser01> Get-EventLog -LogName Security | where-object {$_.EventID -eq 4624}
```

Index	Time	EntryType	Source	InstanceID	Message
2071481	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4624	An account was successfully logged on....
2055169	Feb 12 01:26	SuccessA...	Microsoft-Windows...	4624	An account was successfully logged on....
2055048	Feb 12 01:26	SuccessA...	Microsoft-Windows...	4624	An account was successfully logged on....



```
PS C:\Users\labuser01> Get-EventLog -LogName Security | where-object {$_.EventID -eq 4688}
```

Index	Time	EntryType	Source	InstanceID	Message
2076074	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2074774	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2074768	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2074315	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2071504	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2071148	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2068535	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2068531	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2068219	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2067975	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2067952	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2067385	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2067224	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....
2066776	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4688	A new process has been created....

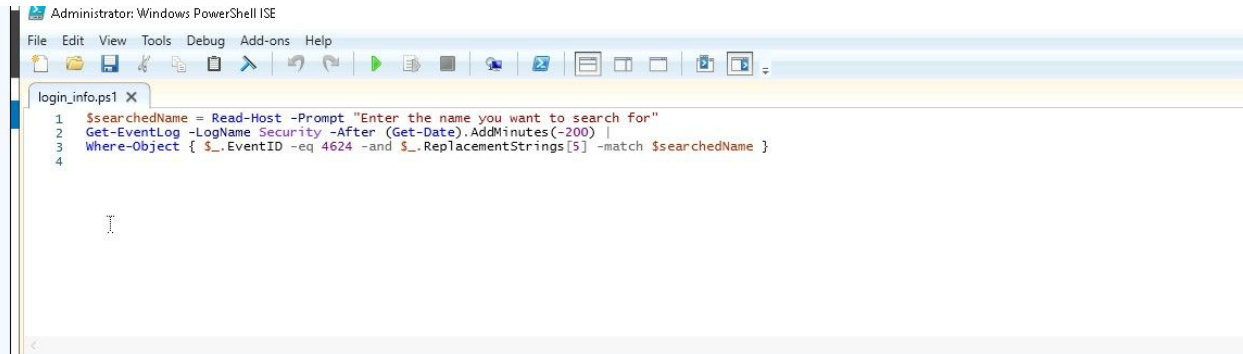


```
PS C:\Users\labuser01> Get-EventLog -LogName Security | where-object {$_.EventID -eq 4689}
```

Index	Time	EntryType	Source	InstanceID	Message
2077000	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2076152	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2074824	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2074823	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2074512	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2074438	Feb 12 01:28	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2069280	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2068491	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2065834	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2065588	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2064950	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2064217	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2063910	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2063896	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2063896	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2062864	Feb 12 01:27	SuccessA...	Microsoft-Windows...	4689	A process has exited....
2058593	Feb 12 01:26	SuccessA...	Microsoft-Windows...	4689	A process has exited....

4.2 Create Login Search Script to track executed processes and commands.

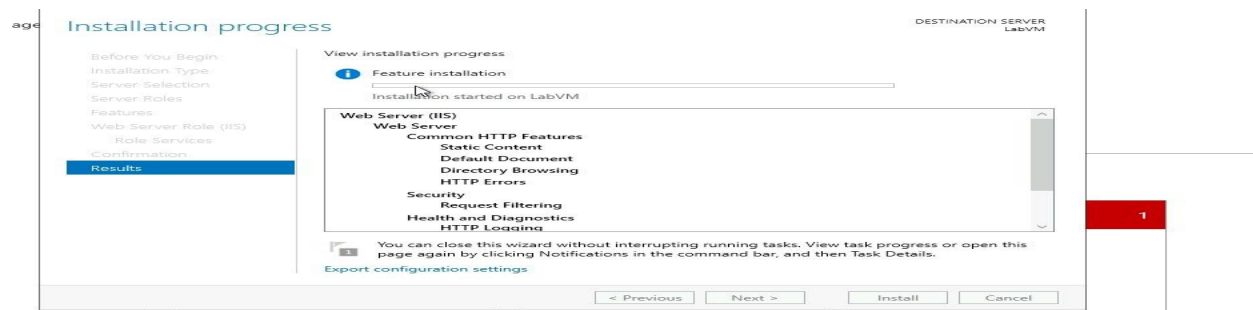
Open PowerShell ISE > Enter the below scripts and save the file as login_info.ps1 on to the Desktop.



Task 5: Automated HTML Security Report Generation

5.1 Install IIS to host security reports in html format

Open Server Manager > Add roles and features > Web Server (IIS) box>Features>Install >Close

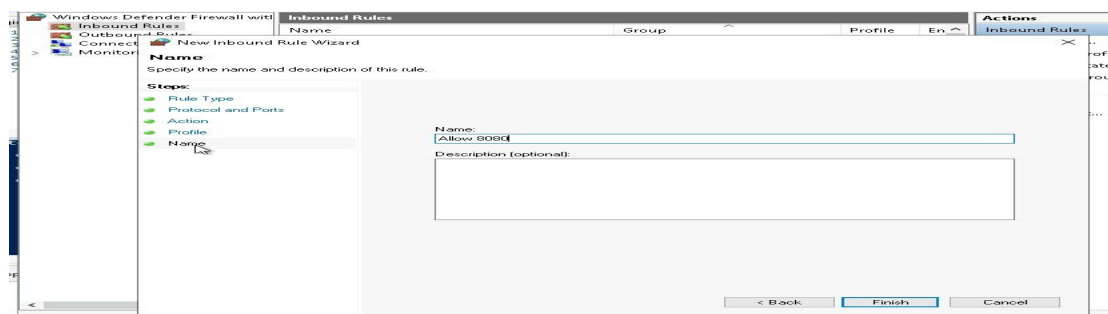
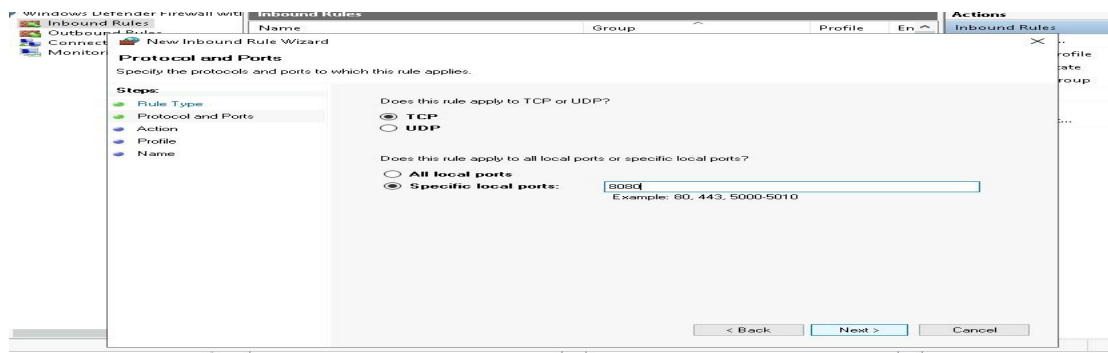


5.2 Create Report Generator Script

Open PowerShell ISE > Enter the below scripts and save as ReportGenerator.ps1 on Desktop.

```
For (; ;)  
{  
    # Get Security events safely and convert to HTML  Get-WinEvent -LogName  
Security | ConvertTo-Html | Out-File "C:\inetpub\wwwroot\security_report.html"  
    # Wait 15 minutes  
    Start-Sleep -Seconds 900  
}
```

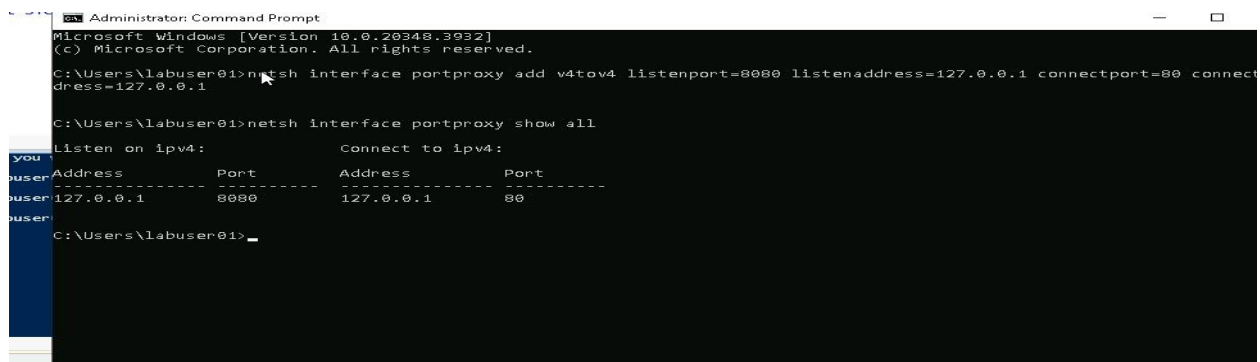
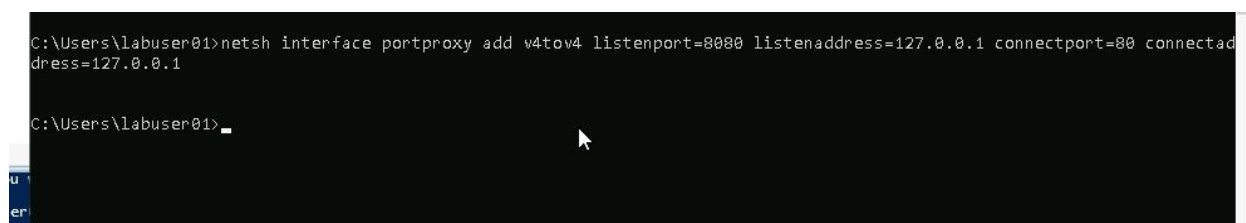
Note: The PowerShell script periodically collects security events and converts them into a structured HTML report stored in: C:\inetpub\wwwroot\security_report.html



6.2 Check Port Forwarding

Open Command Prompt and run the following command:

```
netsh interface portproxy add v4tov4 listenport=8080 listenaddress=127.0.0.1  
connectport=80 connectaddress=127.0.0.1  
netsh interface portproxy show all
```



6.3 Access Reports via port 8080

Open the browser and enter the link http://localhost:8080/security_report.html



EventID	MachineName	Data	Index	Category	CategoryNumber	EntryType	Message	Source	ReplacementString
4660	LabVM	System.Byte[] 3941320 (12801)	12801	SuccessAudit			An object was deleted. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Handle ID: 0x948 Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Transaction ID: (00000000-0000-0000-0000-000000000000)	Microsoft Windows Security Auditing	System.String[]
4663	LabVM	System.Byte[] 3941319 (12801)	12801	SuccessAudit			An attempt was made to access an object. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Object Type: Key Object Name: REGISTRY\MACHINE\SYSTEM\Software\Microsoft\TIP\TestResults\27641370 Handle ID: 0x948 Resource Attributes: - Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Access Request Information: Accesses: %61537 Access Mask: 0x10000	Microsoft Windows Security Auditing	System.String[]
4663	LabVM	System.Byte[] 3941318 (12801)	12801	SuccessAudit			An attempt was made to access an object. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Object Type: Key Object Name: REGISTRY\MACHINE\SYSTEM\Software\Microsoft\TIP\TestResults\27641370 Handle ID: 0x2128 Resource Attributes: - Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Access Request Information: Accesses: %64435 Access Mask: 0x8	Microsoft Windows Security Auditing	System.String[]
4660	LabVM	System.Byte[] 3941317 (12801)	12801	SuccessAudit			An object was deleted. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Handle ID: 0x125c Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Transaction ID: (00000000-0000-0000-0000-000000000000)	Microsoft Windows Security Auditing	System.String[]
4663	LabVM	System.Byte[] 3941316 (12801)	12801	SuccessAudit			An attempt was made to access an object. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Object Type: Key Object Name: REGISTRY\MACHINE\SYSTEM\Software\Microsoft\TIP\TestResults\27641370 (5906C820-845F-4558-99F3-B6DB9CB6ADC) Handle ID: 0x125c Resource Attributes: - Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Access Request Information: Accesses: %61537 Access Mask: 0x10000	Microsoft Windows Security Auditing	System.String[]
4663	LabVM	System.Byte[] 3941315 (12801)	12801	SuccessAudit			An attempt was made to access an object. Subject Security ID: S-1-5-21-292238292-2102740995-1682044382-500 Account Name: labuser01 Account Domain: LabVM Logon ID: 0x13475a Object Object Server Security Object Type: Key Object Name: REGISTRY\MACHINE\SYSTEM\Software\Microsoft\TIP\TestResults\27641370 (5906C820-845F-4558-99F3-B6DB9CB6ADC) Handle ID: 0x948 Resource Attributes: - Process Information: Process ID: 0x640 Process Name: C:\Windows\explorer.exe Access Request Information: Accesses: %64435	Microsoft Windows Security Auditing	System.String[]

Conclusion:

The implemented solution provides TechSecure with:

- Improved detection of unauthorized login attempts
- Visibility into suspicious IP addresses
- Comprehensive logging of commands and processes
- Automated, centralized security reporting
- Reduced response time to potential security incidents

References:

1. Enterprise Infrastructure security course CEP session by Walter
2. Security +Guide to Network Fundamentals

Thank You